Anthony James Padavano

STAFF SOFTWARE ENGINEER & DURABLE AI SYSTEMS

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LINKS

Portfolio · Omni-Dromenon Engine · ORGANVM System · GitHub

PROFILE

Systems architect building durable agentic loops and self-referential state machines for production-grade AI reliability. Sole architect of a 113-repository system across 8 GitHub organizations — 82K files, 23,470 tests, 104 CI/CD pipelines — governing non-deterministic AI outputs through formal state machines and self-healing orchestration. Practitioner of an AI-conductor methodology: human intelligence directing AI-generated volume through structured review gates. Full-Stack Developer (Meta). 18 years professional experience. MFA Creative Writing. LGBTQ+.

TECHNICAL SKILLS

Python · TypeScript · State Machines · Durable Execution · Claude Agent SDK · Multi-Agent Orchestration · Node.js · YAML/JSON Schema · Docker · Redis · GitHub Actions · CI/CD · pytest · Vitest · Zod · Express · WebSocket · GCP / Terraform · Git · pnpm Monorepos · Nginx · React 18

SELECTED PROJECTS

alchemical-synthesizer — Signal-Flow Architecture for Non-Deterministic Logic
Compositional signal-flow system bridging high-level agentic intent and low-level execution reality. Manages.

auto-revision-epistemic-engine — Self-Healing Audit & State Recovery
Formal verification engine for self-referential state management across long-running processes. Automatically.

call-function-ontological — Typed Tool Invocation Framework
Formal ontological layer for structured tool invocation in agentic systems. Maps function calls through typed.

alchemia-ingestvm — High-Throughput Stateful Ingestion Engine
Monorepo-scale ingestion pipeline providing daily health monitoring and signal validation across 8 GitHub.

ai-skills — Modular Agent Capability Registry
Standardized framework for AI agent tool invocation with typed inputs, validation gates, and formal skill.

EXPERIENCE

Staff Software Engineer & Durable AI Systems — 2020 — Present
ORGANVM

- **Durable State Machines:** Designed 113-repository system governed by formal promotion pipelines (LOCAL.
- **Multi-Agent Orchestration:** Built agentic frameworks with stateful retries, memory management, and.
- **Signal-Flow Architecture:** Compositional synthesizer bridging non-deterministic AI outputs to.
- **CI/CD Infrastructure:** 128 GitHub Actions workflows (18 reusable templates), 104 CI/CD pipelines,.

Instructor — Composition & Writing — 2015 — Present
Miami Dade College, Keiser University, Nova Southeastern, FAU, Young Harris, Tarrant County, Broward, Palm Beach State

- 100+ courses across 8 institutions; 2,000+ students. Technical communication, systems-thinking pedagogy, and structured.

Digital Marketing Manager — 2023 — 2024
Miami Dade College Foundation (nonprofit)

- Rebuilt foundation website; designed data-driven fundraising campaigns increasing donor engagement by 32%. End-to-end.

Multimedia Specialist — 2011 — 2020
AJP Media Arts — clients incl. Boca Ballet Theatre, Forward Funding, Arts in Boca

- Video, audio, and visual production pipelines for nonprofits and arts organizations; 17.5M+ views, $2M in fundraising.

EDUCATION

M.F.A. Creative Writing, Florida Atlantic University — 2015 — 2018
B.A. English Literature, CUNY College of Staten Island — 2010 — 2014

CERTIFICATIONS

Full-Stack Developer, Meta (2024) · UX Design, Google (2023) · Digital Marketing, Google (2023) · Project Management, Google (2023)